TOPIC 5

Feature Selection Techniques: Explained

The original class slides, with a plain explanation under each one

> **NOTE** **How to read this document**
>
> - Every original slide is kept exactly as it was, with its slide number under it
> - Under each slide (or small set of slides) you get a plain explanation: **what you are looking at** and **what it means**
> - The slides come from a recorded class, so a few look like screenshots of a video: that is normal

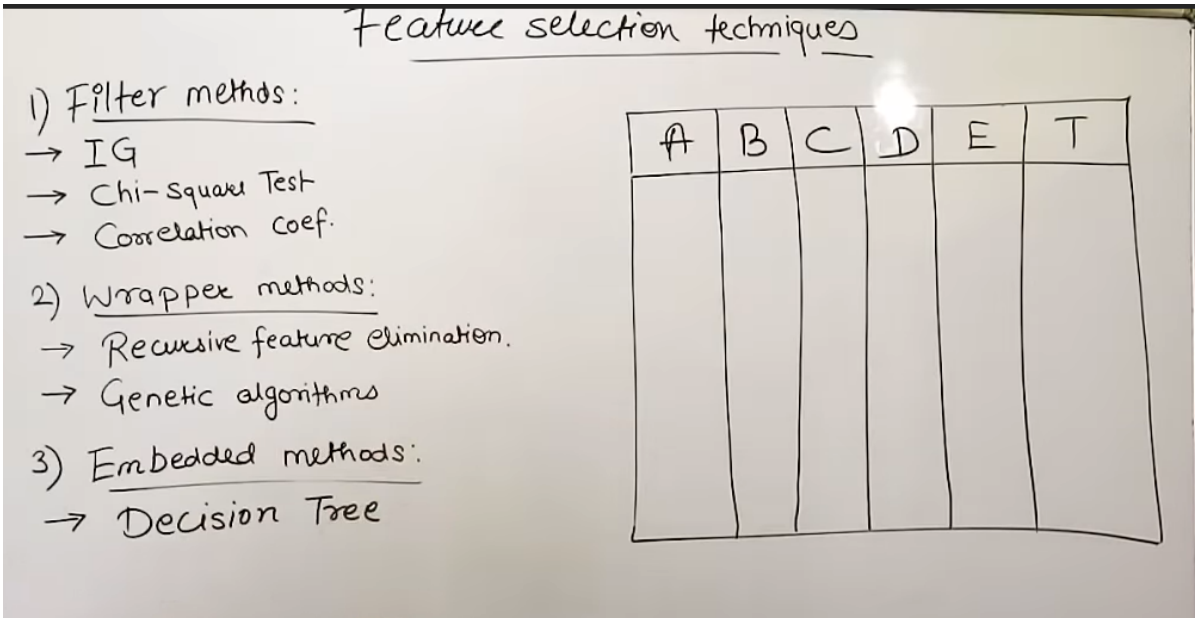
Feature selection means choosing which columns (features) of a data table are worth giving to a model. These 11 slides are whiteboard pages: first a list of the techniques, then the **chi-square test** worked on one example, then drawings showing how subsets of columns are tried with a model.

Throughout, the table with columns **A, B, C, D, E** and **T** is the data: A to E are the **input features** and **T** is the **target** (the thing to predict). Each lettered circle **M** is a **model**.

Words you will meet

Word	Plain meaning
Feature	One input column of the data (A, B, C, D or E here)
Target (T)	The column we want to predict
Filter method	Scores each feature on its own with a statistic, with no model involved
Wrapper method	Trains a model on different subsets of features and keeps the best subset
Embedded method	The model chooses features while it is being trained (for example a decision tree)
Chi-square (χ^2)	A statistic that compares observed counts with expected counts
Observed (O) / Expected (E)	What really happened / what we would expect if nothing special were going on
Degrees of freedom (df)	A number from the table size that is needed to look up the critical value

Slide 1: The map of the whole topic

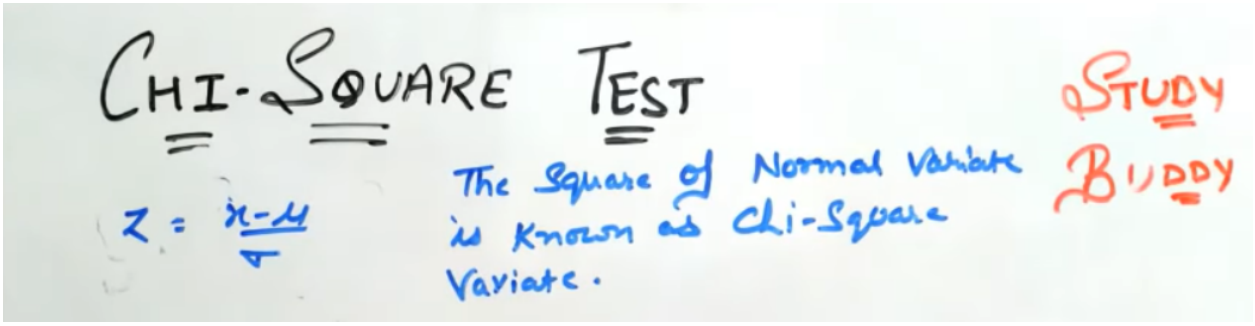


Slide 1

The whiteboard lists the three families of feature selection and an example of each:

- **1) Filter methods:** IG (Information Gain), the Chi-Square test, and the Correlation coefficient. They score each feature separately.
- **2) Wrapper methods:** Recursive Feature Elimination and Genetic algorithms. They try groups of features with a model.
- **3) Embedded methods:** Decision Tree. The model itself reveals which features matter.

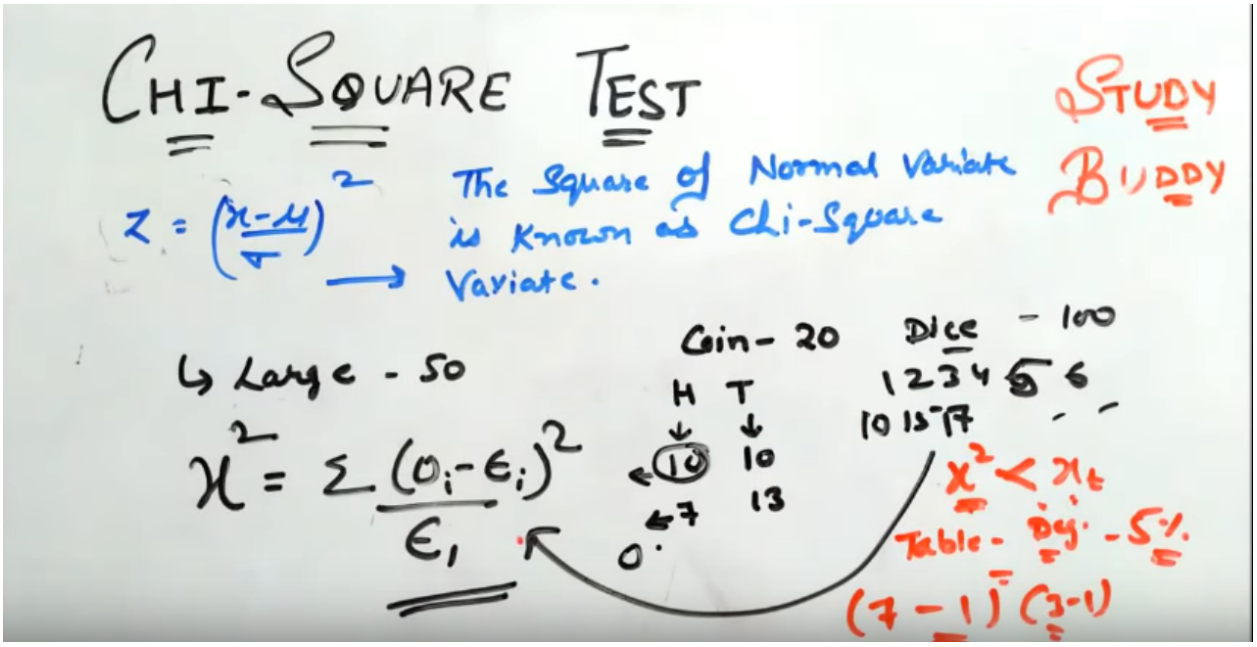
Slide 2: Chi-square: where the name comes from



Slide 2

This slide (from the 'Study Buddy' lesson) starts with the **z-score**, z = (x − μ)/σ: how many standard deviations a value is from the average. It says: **the square of a normal variate is known as a chi-square variate.** In plain words: if you take z-scores and square them, add them, you get a chi-square number.

Slide 3: The chi-square formula

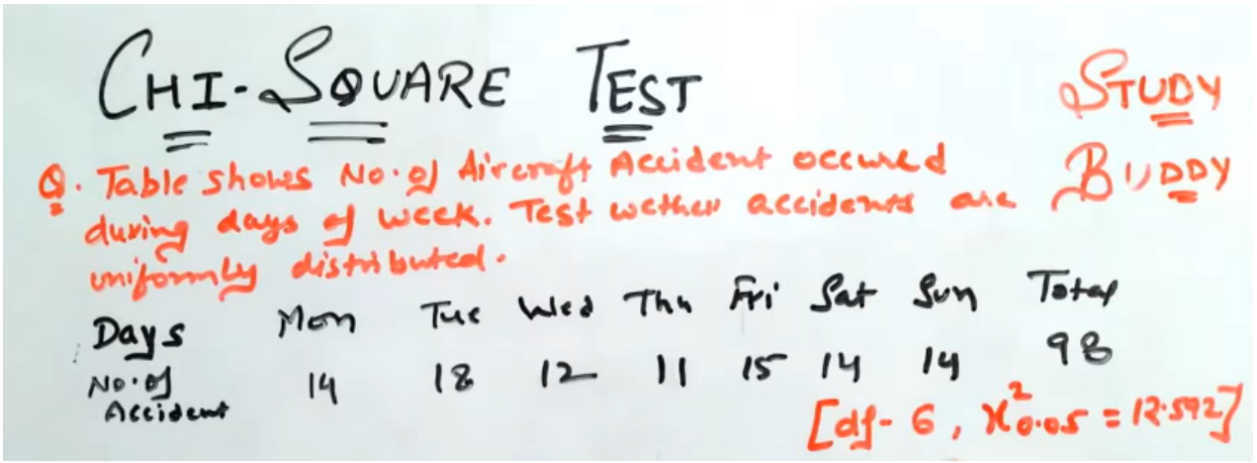


Slide 3

Squaring the z-score gives the **chi-square** formula on the left: **$\chi^2 = \Sigma\,(O_i - E_i)^2 / E_i$**. O is the **observed** count and E is the **expected** count. The board also shows side examples (a coin tossed 20 times, a dice rolled 100 times) of observed versus expected counts. The rule at the right: if the calculated χ^2 is **smaller than** the table value (at the 5% level) we accept the hypothesis; if larger, we reject it.

$$\chi^2 = \sum_i \frac{(O_i - E_i)^2}{E_i}$$

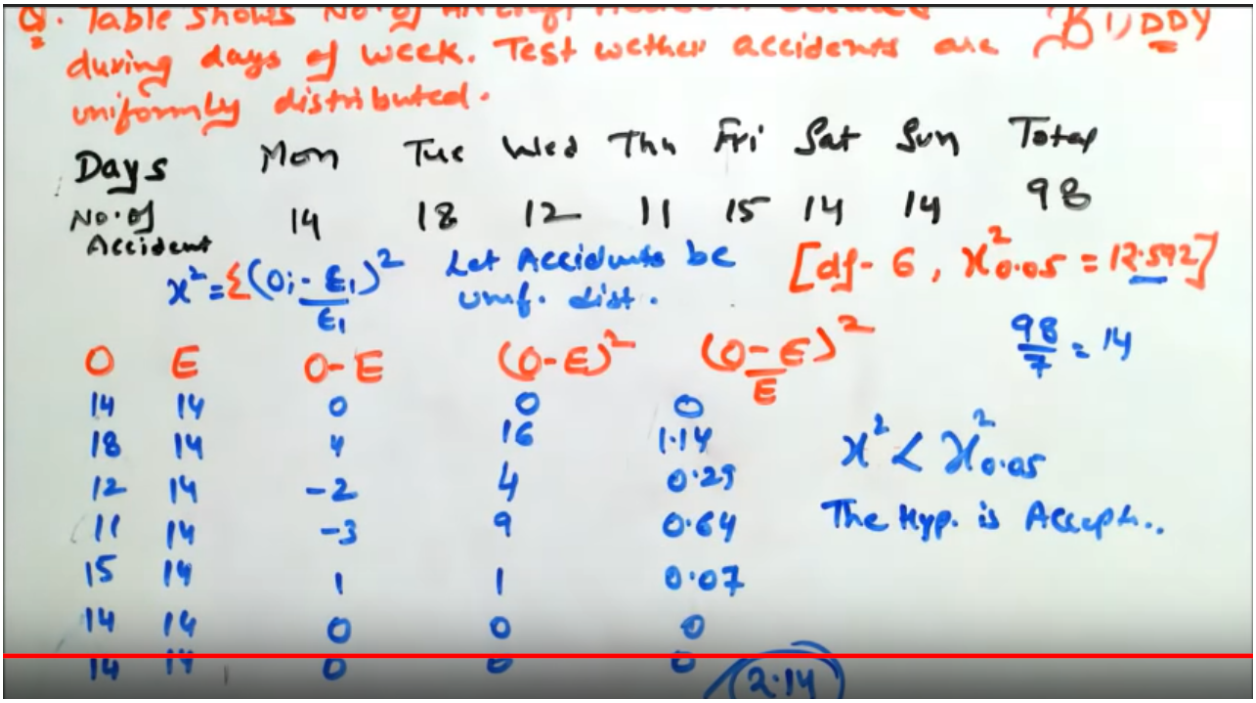
Slide 4: A worked question: aircraft accidents



Slide 4

Question: the table shows the **number of aircraft accidents on each day of the week**. Test whether accidents are **uniformly distributed** (equally likely every day). The numbers are Mon 14, Tue 18, Wed 12, Thu 11, Fri 15, Sat 14, Sun 14, with a total of 98. Written in the corner: **df = 6** (7 days minus 1) and the **table value $\chi^2_{0.05} = 12.592$**.

Slide 5: The working

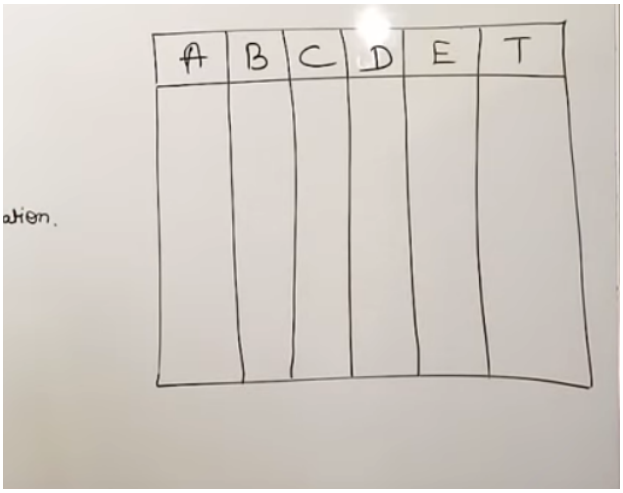


Slide 5

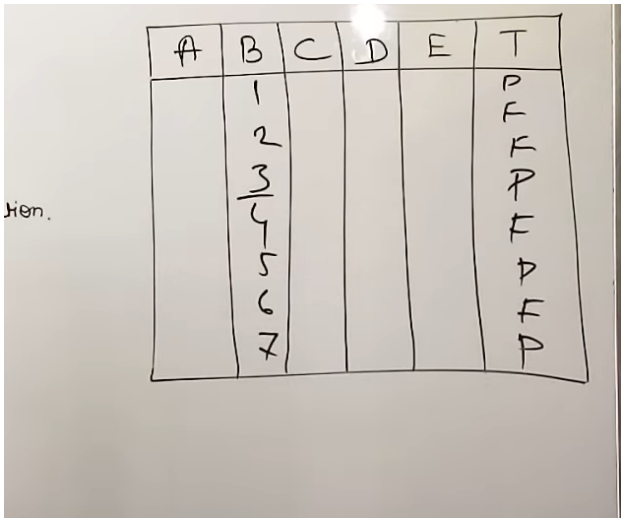
If accidents are spread evenly, each day expects 98 ÷ 7 = **14**. For every day the board computes O − E, then $(O - E)^2$, then $(O - E)^2/E$:

- Mon 14: 0; Tue 18: 4 → 16 → 1.14; Wed 12: −2 → 4 → 0.29
- Thu 11: −3 → 9 → 0.64; Fri 15: 1 → 1 → 0.07; Sat and Sun 14: 0
- Add up all the last numbers: **χ^2 = 2.14**
- Since 2.14 is **less than** 12.592, **the hypothesis is accepted**: accidents are evenly spread over the week (the differences are just chance)

Slides 6 to 7: Back to the data table



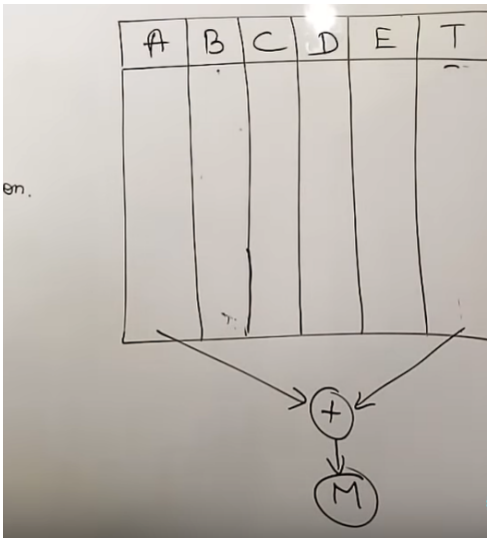
Slide 6



Slide 7

The teacher returns to the table of features A, B, C, D, E and target T. Slide 7 numbers the rows 1 to 7 under feature **B** and writes a short label in column **T** for each row (letters such as P and F): the target holds a category for every row, which is what the chi-square test compares each feature against.

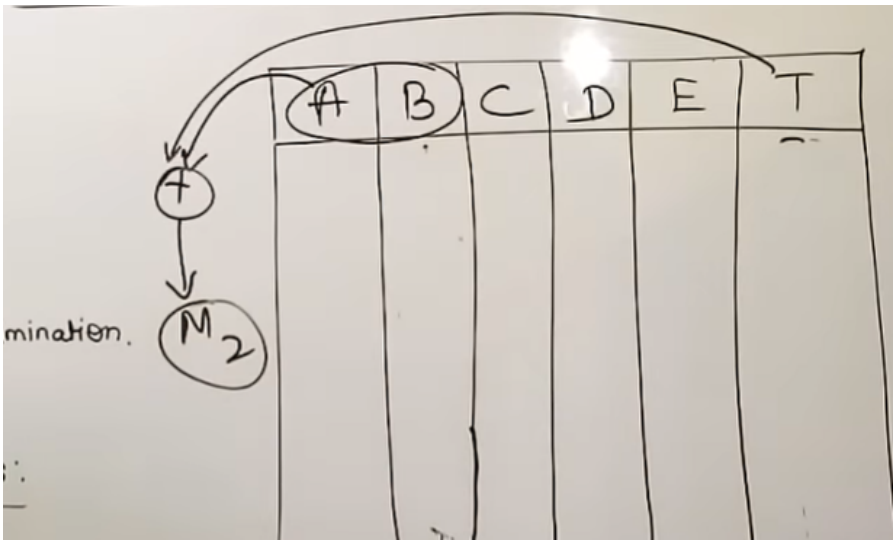
Slide 8: Wrapper idea, step 1: model with all features



Slide 8

All the columns, features plus target, are fed into a **model, M** (the plus sign means 'combine and train'). This is the starting point: a model that uses every feature. Its score is the number we will compare other models against.

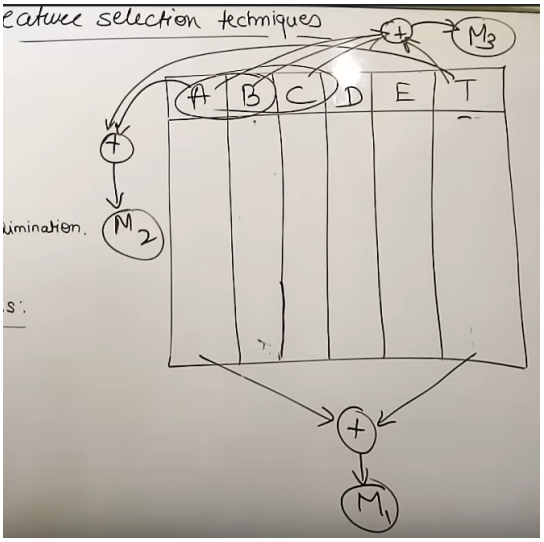
Slide 9: Wrapper idea, step 2: a model with fewer features



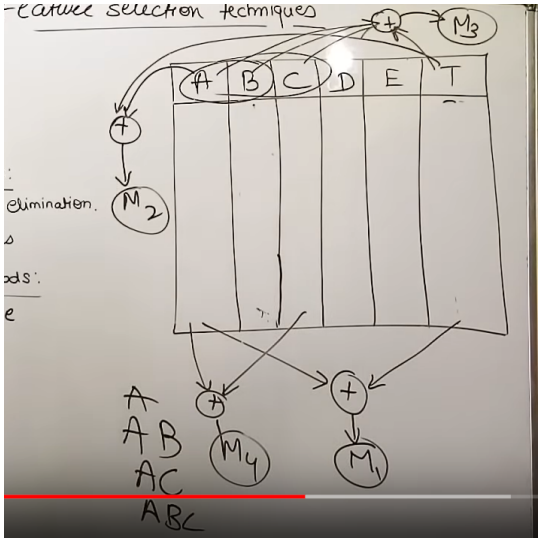
Slide 9

Now only features **A and B** (circled) together with T are fed into a second model, **M_2**. By comparing M_2 with the full model we can see whether dropping C, D and E makes the result worse or not. If it does not, those columns were not needed.

Slides 10 to 11: Wrapper idea, step 3: try many subsets



Slide 10



Slide 11

More subsets are tried, each with its own model: **A, B, C → M₃** and other combinations, next to the full model M₁. Slide 11 writes down the subsets being tried: **A**, **A B**, **A C**, **A B C**, each with its own model (M₄ and so on). The subset whose model scores best is kept.

- This is how a **wrapper method** works: it *wraps* a model around the search for the best set of features
- It is accurate but slow, because many models must be trained